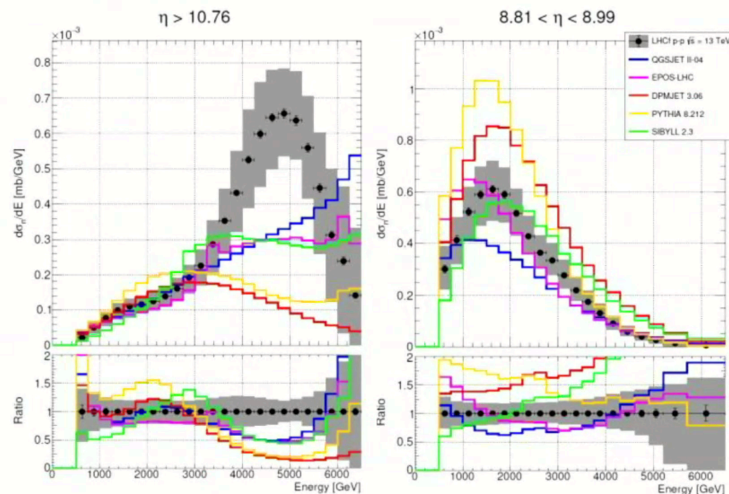


- Questions
 - Heiko
 - Neutrons AND K-long? Is it possible two distinguish them
 - Machine Learning
 - p0 production - muon puzzel?
 - Neutron production is more crucial
 - Baryonic
 - Both contributions are important
 - Non trivial
 - There is actually a part missing - pion interaction

LHCf and hadronic interaction models: Neutron spectra



- Masterstudent has to measures response in SND - SND is in the other direction - what can we learn from a measurement on the other side

- Fragmentation is very missmodeled
 - What could be learned from the photon production
 - Compare the spectra from O-p collisions with p-p collisions
 - In the models what is the primary source of photons?
 - Proton bremsstrahlung (completely unmeasured at these energies)
 - Could be a source of dark photon production
 - single photons from p0
- Jonas
 - Why do you only look at neutral particles
 - AFP experiment is measuring protons
 - no experiment that can measure f.e. pi- particles
 - Why not nitrogen?* - more difficult to prepare and accelerate - they are very similar
- Collision with the CMB? UHECR from their reference frame the cosmic microwave background looks like very high energetic photons
- Detection efficiency is higher for photons than for hadrons
- Energy and Position at the same time? In ATLAS they do not use silicon layers
 - CMS DHcal - they are working on this
 - Currently a money question
- Elisabeth
 - What is the cross section of the oxygen proton collision
 - How does it differ from the proton collision
 - Not sure which momentum transfer we look for

